

# Root-Cause Report - Sample Deliverable

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## Root-Cause Report - sample one-pager

\*The named deliverable of the debug gig. Real bug, anonymized. Every fix-tier order ships with a report in exactly this shape; the Basic "Reproduce + Diagnosis" tier ships everything except the fix + test sections.\*

### Symptom (client's words)

- "My bot stopped processing anything. It runs without errors, the logs look normal, but it silently skips every item as 'wrong project'."

No crash. No traceback. A system that looks alive but does nothing.

### Reproduction

1. Ran the script against a copy of the database - reproduced on the first attempt. The failure is deterministic, not intermittent.
2. Minimal repro: one session row + the `is-mine` check returns `False` for every session regardless of the real project.

### Root cause

The script read SQLite rows by positional column index:

```
cwd = row[4]    # assumed column #4 was cwd
```

A `last\_activity\_at` column was later added in the middle of the schema. The `SELECT` list was updated, but the reader still walked the tuple by position - so index 4 now held a timestamp, not the project directory. Every session looked like it belonged to a different project, and the "is this mine?" check silently rejected all of them.

The tell: it appeared only after a schema change, and it never threw - it just mis-read data.

### Why it stayed hidden

- No assertion or test tied the row reader to the schema.
- The failure mode was "wrong answer," not an exception - logs looked clean.

### The fix (Standard tier) + class fix (Premium tier)

Read by column name, never by position:

```
rec = dict(zip(select, row))    # named, order-independent
cwd = rec.get("cwd") or ""
```

- Order-independence: keyed by name, a reordered schema can't shift data.
- Schema-graceful: a missing column is absent (`None`), never shifted.
- Class fix: this removes the entire positional-read bug class across the codebase - every other positional read got the same treatment.

## Regression test (proves the fix)

```
# columns deliberately shuffled; reads must still return cwd correctly
assert load_session(db, sid)["cwd"] == "/expected/path"
```

If anyone reorders the schema again, the test fails loudly instead of the bot silently skipping everything.

## Warranty

The fix carries a 14-day warranty: if this bug reappears in that window, the fix is free.

\*Representative sample: real bug, real code, edited only to remove project-specific context. Gallery-sized rendering available in ``../assets/gallery-root-cause-report.png``.\*